

Waldenström macroglobulinemia

Lymphoplasmacytic lymphoma with an IgM paraprotein · **MYD88 L265P** in ~90%+ · treat symptoms and IgM complications, not the IgM number · genotype (MYD88/CXCR4) steers BTKi vs chemoimmunotherapy

Compiled from primary trial reports and consensus guidelines (NCCN; IWWM-2, IWWM-11 and IWWM-12 consensus panels) verified against PubMed, September 2026.

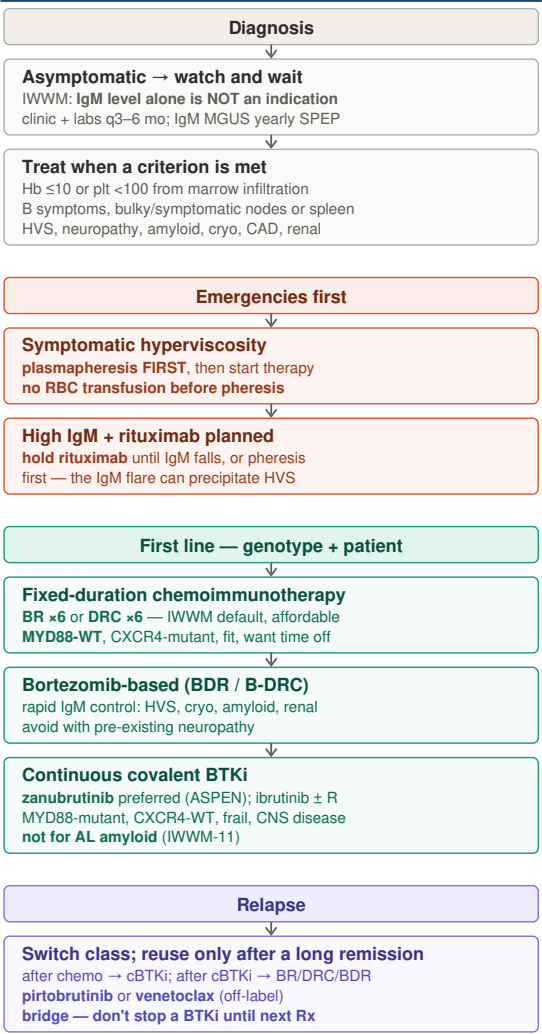
DEFINITIONS AND THE IGM DIFFERENTIAL

Entity	Defining features
WM	Lymphoplasmacytic lymphoma (LPL) in the marrow + an IgM monoclonal protein of any size. IWWM-2 and IWWM-11: no IgM or marrow-% cutoff separates WM from IgM MGUS — the diagnosis rests on marrow histology + immunophenotype
IgM MGUS	IgM M-protein, no LPL on marrow, no symptoms. Progresses to lymphoma/WM at ~1.5%/yr. IWWM-11 splits it: plasma-cell type (MYD88 wild-type — myeloma/AL lineage) vs B-cell type (may be MYD88-mutant)
Asymptomatic (smoldering) WM	Meets WM criteria, no treatment indication → observe . 72% progress (median follow-up 7.8 yr). Progression risk: IgM ≥4,500 mg/dL, marrow ≥70%, β2M ≥4 mg/L, albumin ≤3.5 ; MYD88 wild-type HR 2.7
IgM myeloma	CD138+ plasma cells without a B-cell/LPL component, t(11;14), MYD88 wild-type, lytic lesions. Treat as myeloma — not WM
Marginal zone	Can make IgM and show plasmacytic differentiation. MYD88 L265P absent or rare — the most useful discriminator
IgM-related disorder	Symptoms from the IgM itself (neuropathy, cryo, CAD, amyloid) without overt lymphoma — still may need clone-directed therapy

WORKUP

- **SPEP + immunofixation, quantitative IgM** (SPEP M-spike and nephelometric IgM are not interchangeable — follow one method), free light chains, β2-microglobulin, albumin, LDH, CBC, CMP
- **Marrow biopsy:** intertrabecular LPL infiltrate; surface IgM+, CD19+, CD20+, CD22+, with CD138+ plasma cells; usually CD5– CD10– CD23– (variants do not exclude WM)
- **Molecular on marrow before any therapy** (IWWM-11): **AS-PCR for MYD88 L265P and CXCR4 S338X on whole marrow**; CXCR4 + TP53 sequencing and 6q/17p FISH on CD19-selected cells
- **CT chest/abdomen/pelvis** for nodes/spleen. **Serum viscosity + fundoscopy** if IgM is high or any HVS symptom
- Targeted: **cryoglobulins** (draw and transport at 37°C), **DAT + cold agglutinin titer, anti-MAG + EMG** if neuropathy, **NT-proBNP/troponin, urine protein, fat pad** if amyloid suspected, **MRI brain/spine + CSF** if CNS symptoms
- **Prognosis:** IPSSWM — age >65, Hb ≤11.5, plt ≤100, β2M >3 mg/L, M-protein >7 g/dL; 5-yr OS **87% / 68% / 36%** (low/int/high). **Not a treatment trigger**

TREATMENT PATHWAY



IGM-MEDIATED COMPLICATIONS — EACH MANAGED DIFFERENTLY

Syndrome	Recognize → manage
Hyperviscosity <i>large pentameric IgM</i>	Blurred vision, headache, epistaxis/oronasal bleeding, dizziness, confusion, heart failure. Fundoscopy: dilated “sausage” veins, hemorrhages, papilledema. Viscosity poorly predicts symptoms per patient. Plasmapheresis first (IgM is ~80% intravascular, so it clears efficiently), then cytoreduce — pheresis alone is a bridge. Avoid RBC transfusion until pheresed
Anti-MAG neuropathy	Slowly progressive, distal, symmetric sensory-predominant demyelinating neuropathy with ataxia/tremor. EMG + anti-MAG titer; joint heme-neuro care. Rituximab-based therapy ; stabilization is the realistic goal. Avoid bortezomib (and vincristine)
Cryoglobulinemia <i>type I or II</i>	Raynaud's, acral necrosis, purpura, vasculitic neuropathy, GN. Draw at 37°C or it is missed. Keep warm; plasmapheresis for severe ; rapid cyto reduction (BR, BDR)
Cold agglutinin disease	Chronic hemolysis with cold acrocyanosis; DAT C3d+, IgMk titer. Keep warm, warm IV fluids/blood. Rituximab-based (BR most active); sutimlimab blocks C1s for hemolysis
AL amyloidosis	Lambda light chain, cardiac/renal/nerve. Treat as AL: bortezomib- or bendamustine-based; daratumumab. cBTKi not recommended (IWWM-11)
Bing-Neel <i>CNS WM</i>	Headache, cognitive change, cranial nerve palsies, gait. MRI brain + spine with gadolinium, CSF cytology/flow + MYD88 PCR. BTKi cross the BBB: ibrutinib (n=28: 85% symptom improvement, 2-yr EFS 80%); IWWM-12 made zanubrutinib a standard. Alternatives: HD-MTX / HD-araC, fludarabine

GENOTYPE DRIVES BTKI CHOICE

Lesion	Why it matters
MYD88 L265P ~90%+	Drives NF-κB via IRAK/BTK. Supports the diagnosis (rare in MZL, absent in myeloma). MYD88 wild-type: weaker BTKi response (ibrutinib major response 29%), worse OS — favor chemoimmunotherapy
CXCR4 WHIM-like ~30–40%	Nonsense/frameshift C-tail mutations; higher IgM + marrow burden, more hyperviscosity. Slower, shallower BTKi responses: TN ibrutinib major response 71% vs 94%, VGPR 7% vs 31%, time to major response 7.3 vs 1.8 mo
TP53, del6q, del17p	Adverse; IWWM-11 counts TP53 and 6q/17p among essential pre-treatment tests

FIRST-LINE OPTIONS — THE EVIDENCE

Regimen	Evidence and the catch
Bendamustine-rituximab	IWWM/NCCN preferred chemoimmunotherapy. Randomized evidence is StiL NHL1 (BR beat R-CHOP in indolent NHL, WM a small subgroup). Rituximab maintenance after BR: no benefit overall (StiL NHL7-MAINTAIN)
DRC	Phase 2, n=72: ORR 83% (CR 7%), 2-yr PFS 67%; grade 3–4 neutropenia only 9%. Slow: median time to response 4.1 mo . ECWM-1: adding bortezomib (B-DRC) did not improve PFS (24-mo 80.6% vs 72.8%, P=0.32) but responses came faster (3.0 vs 5.5 mo)
BDR	Weekly-bortezomib EMN regimen: ORR 85% , median PFS 42 mo ; neuropathy 46% (grade ≥3 7%). Bortezomib lead-in blunts rituximab flare. Zoster prophylaxis mandatory
Ibrutinib ± rituximab	INNOVATE (phase 3, n=150): IR vs placebo-R — final median PFS NR vs 20.3 mo , HR 0.25; benefit regardless of MYD88/CXCR4 . Adding ibrutinib cut rituximab IgM flare (8% vs 47%). Grade ≥3 AF 12% vs 1% , HTN 13% vs 4%
Zanubrutinib	ASPEN (phase 3, n=201 MYD88-mutant): primary endpoint VGPR+CR 28% vs 19% , P=0.09 — not met . Final (44 mo): VGPR+CR 36% vs 25%; CXCR4-mutant 21% vs 10%. Less AF (7.9% vs 23.5%) , HTN, bleeding, diarrhea; more neutropenia. FDA-approved Aug 2021
Acalabrutinib	Single-arm phase 2, n=106: ORR 93% TN and R/R; grade 3–4 AF 1%. Not FDA-approved for WM
Ibrutinib + venetoclax	Fixed-duration TN trial: VGPR 42% but ventricular arrhythmia 9% incl. 2 deaths → stopped early . Not a standard

FIXED-DURATION VS CONTINUOUS — HOW TO CHOOSE

- **Chemoimmunotherapy**: finite (~6 mo), then time off therapy; myelosuppression, infections; the **better fit for MYD88 wild-type** and a reasonable one for CXCR4-mutant disease
- **cBTKi**: oral, rapid Hb recovery, **indefinite** — cost, cumulative AF/HTN/bleeding, and **IgM rebound if stopped**. Best in MYD88-mutant/CXCR4-WT; zanubrutinib over ibrutinib for tolerability
- **IgM lags the marrow**: IgM can plateau or rise transiently (rituximab flare; **bortezomib/rituximab-era discordance**) — do not call progression on IgM alone; confirm, check marrow/Hb/nodes. BR/DRC responses deepen for months after the last cycle

RELAPSED / REFRACTORY

- **Durable first remission → the same regimen can be reused**; early relapse or refractory → switch class
- **Ibrutinib withdrawal**: IgM rebound (≥25%) in **73%** of discontinuations, half within 4 weeks, some with HVS; salvage works better if started within 2 weeks. **Continue the BTKi until the next therapy starts**; hold only ~3–7 days peri-procedure
- **Pirtobrutinib** (non-covalent; binds C481-mutant BTK): BRUIN WM n=80 (79% cBTKi-exposed) — ORR **82.5%** (81% post-cBTKi), major response ~72%. **Not FDA-approved for WM** (MCL/CLL only) — single-arm, off-label
- **Venetoclax** 200→800 mg daily ×2 yr: ORR 84%, major 81%, median PFS **30 mo**; slower after BTKi (4.5 vs 1.4 mo); **CXCR4 did not matter**. Neutropenia 45%. Off-label
- Also: bortezomib/carfilzomib-based, BR after BTKi, clinical trials (BTK degraders, CXCR4 antagonists, bispecifics). **Autologous SCT** for select chemosensitive young patients. Transformation to DLBCL → treat as DLBCL

RESPONSE (IWWM-6 / SIMPLIFIED IWWM-11)

CR	Normal IgM, negative immunofixation , no marrow disease, resolved nodes/spleen
VGPR	≥90% IgM reduction
PR	50–89% IgM reduction · major response = PR or better
MR	25–49% IgM reduction (counts in "overall" response)
PD	≥25% IgM rise from nadir (confirmed) or new disease/complication

IWWM-11 lets PR/VGPR be called on serum IgM alone, with guidance for flare/rebound. **ORR in WM trials includes minor responses** — compare major response rates across studies.

REGIMEN DOSING

Regimen	Dosing
BR	Bendamustine 90 mg/m² d1–2 + rituximab 375 mg/m² d1, q28d ×6
DRC	Dexamethasone 20 mg IV + rituximab 375 mg/m² d1 + cyclophosphamide 100 mg/m² PO BID d1–5 , q21d ×6
BDR EMN	Cycle 1 bortezomib alone 1.3 mg/m² d1,4,8,11; then 1.6 mg/m² weekly d1,8,15,22 q35d ×4 ; dex 40 mg + rituximab 375 in cycles 2 and 5. SC bortezomib in practice
Zanubrutinib	160 mg BID or 320 mg daily , continuous
Ibrutinib ± R	420 mg daily , continuous; iINNOVATE rituximab 375 mg/m² weekly, weeks 1–4 and 17–20
Acalabrutinib	100 mg BID, continuous
Pirtobrutinib	200 mg daily, continuous
Venetoclax	Ramp 200 → 800 mg daily , up to 2 yr; TLS precautions

MONITORING AND PROPHYLAXIS

- **HBsAg + anti-HBc before any anti-CD20** — entecavir if core-Ab positive. **Acyclovir/valacyclovir with bortezomib** (zoster in 4 of the first 7 BDR patients) and commonly with BR and BTKi
- **BTKi**: baseline ECG and BP; ask about palpitations; **treat HTN**. AF → rate control, weigh anticoagulation with bleeding; switch ibrutinib → zanubrutinib. **No warfarin; avoid fish oil/vit E**; hold ~3–7 days around surgery. Strong CYP3A inhibitors → dose-adjust
- **Bortezomib**: neuropathy check every cycle; **weekly + SC** dosing lowers it; dose-reduce at grade 2 with pain
- **Rituximab**: IgM before each dose; **hold or pre-pherese if IgM is high** (commonly ≥4,000 mg/dL); a rising IgM in the first months is often flare, not progression
- IgM + CBC q1–3 mo on therapy; SPEP/IgM q3–6 mo in remission. **Same lab, same method**

KEY TRIALS

INNOVATE	Ibrutinib-R vs placebo-R — median PFS NR vs 20.3 mo, HR 0.25, genotype-independent
ASPEN	Zanubrutinib vs ibrutinib — VGPR+CR primary not met (28 vs 19%); far less AF
ECWM-1	B-DRC vs DRC — no PFS gain; faster response
DRC (Dimopoulos)	Phase 2 — ORR 83%, 2-yr PFS 67%
BDR (EMN)	Phase 2 — ORR 85%, median PFS 42 mo, weekly bortezomib
Ibrutinib TN / R/R	Phase 2 — CXCR4 and MYD88 genotype set depth and speed
BRUIN WM	Pirtobrutinib R/R — ORR 82.5%, active post-cBTKi
Venetoclax (Castillo)	R/R — major response 81%, median PFS 30 mo